

Tool	Why choose it?
Selenium IDE	To learn about concepts on automated testing and Selenium, including Selenium commands such as type, open, clickAndWait, assert, verify, etc. Locators such as ID, name, xpath, CSS selector, etc. Executing customised JavaScript code using run script Exporting test cases in various formats To create tests with little or no prior knowledge in programming To create simple test cases and test suites that you can export later to RC or WebDriver To test a Web application against Firefox only
Selenium RC	To design a test using a more expressive language than Selenium To run your test against different browsers (except HTML Unit) on different operating systems To deploy your tests across multiple environments using Selenium Grid To test your application against a new browser that supports JavaScript To test Web applications with complex Ajax-based scenarios
Web-Driver	To use a certain programming language in designing your test case To test applications that are rich in Ajax-based functionalities To execute tests on the HTML Unit browser To create customised test results
Selenium Grid	To run your Selenium RC scripts in multiple browsers and operating systems simultaneously To run a huge test suite, that needs to complete within the quickest time possible

Table 1: Choosing the right tool

	Selenium IDE	Selenium RC	WebDriver
Browser support	Mozilla Firefox	Mozilla Firefox, Internet Explorer, Google Chrome, Safari, Opera, Konqueror, Others	Internet Explorer versions 6 to 9, both 32- and 64-bit Firefox 3.0, 3.5, 3.6, 4.0, 5.0, 6, 7 and above (current version is 16.0.1) Google Chrome 12.0.712.0 and above (current version is 22.0.1229.94 m) Opera 11.5 and above (current version is 12.02) Android - 2.3 and above for phones and tablets (devices and emulators) iOS 3+ for phones (devices and emulators) and 3.2+ for tablets (devices and emulators) HTML Unit 2.9 and above (current version is 2.10)
Operating system	Windows, Mac OS X, Linux	Windows, Mac OS X, Linux, Solaris	All operating systems where the browsers given above can run

Table 2: Browser and environment support

- Has built-in help and a test results reporting module
- Provides support for extensions
The cons are:
 - Available only in Firefox
 - Designed only to create the prototypes of tests
 - No support for iteration and conditional operations
 - Test execution is slow compared to that of Selenium RC and WebDriver

Selenium Remote Control (RC)

Selenium RC allows you to write automated Web application tests in compliant programming languages. It offers programming language options like Java, Ruby, Python, Perl, PHP or .NET.

The pros of this tool are:

- Cross-browser and cross-platform
- Can perform looping and conditional operations
- Can support data-driven testing
- Has a mature and complete API
- Can readily support new browsers
- Faster execution than the IDE
The cons are:
 - Installation is more complicated than with Selenium IDE
 - Requires users to have programming knowledge
 - Needs the Selenium RC server to be running
 - API contains redundant and confusing commands
 - Browser interaction is less realistic
 - Inconsistent results and uses JavaScript
 - Slower execution time than WebDriver

Selenium WebDriver

Selenium WebDriver itself is more robust than both RC and IDE. It uses a more stable approach to automate browser actions. Rather than calling any other external JavaScript, WebDriver directly interacts with the browser and its components. It supports the same programming languages as